Paul He

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Education

University of Toronto — September 2021 - May 2025
Honours Bachelor of Science in Computer Science — *Toronto, Canada*

- Enrolled in Arts & Science Co-Op program
- Dean's List
- **Relevant Coursework:** Introduction to Machine Learning

Swiss Federal Institute of Technology (ETH Zürich) — September 2023 - August 2024
Exchange Year — *Zürich, Switzerland*

- **Relevant Coursework:** Visual Computing, Natural Language Processing (Graduate), Probabilistic Artificial Intelligence (Graduate), Machine Perception (Graduate), Large Language Models (Graduate), Advanced Formal Language Theory (Graduate)

Research

Joint Chinese Word Segmentation (CWS) and Part-of-Speech Tagging (POS) — October 2023 – January 2024
Supervisors: Tianyu Liu, Prof. Dr. Ryan Cotterell — *ETH Zürich*

- Engaged fully in the research lifecycle: conducted a literature review, developed the model, performed experimentation, documented findings, and authored a comprehensive research paper.
- Implemented a **transformer** model using **PyTorch**, beam-search for generation.
- Achieved leading-edge results, demonstrating the model's efficiency with a validation accuracy of **96.29%** for CWS, **93.37%** for POS tagging, and **91.99%** for joint CWS and POS tagging after just 10 epochs.

Projects

Mental Health Support Bot | *Python, beautifulsoup4, FastAPI* — October 2023

- Led the development of simple mental health chatbot web-app tailored for ETH Computer Science students, assisting team members in debugging **under 48 hours** (Hackathon).
- Performed **web scraping** to fine-tuned pre-trained language model.
- Implemented a real-time chat feature FastAPI to host our chat platform and model on a server, allowing multiple chat's with the bot at once, and also recalls previous conversations

Game Matchup Estimator | *Python, Pytorch, Numpy, scikit-learn, Flask, HTML, CSS* — May 2023 - September 2023

- Integrated the Clash Royale API to retrieve real-time game data, generating up to 130,000 labeled data points.
- Developed a neural network model with **PyTorch** to analyze large-scale data and accurately predict match outcomes with 73% accuracy.
- Utilized advanced techniques such as **feature engineering, regularization, and hyperparameter tuning** to enhance the model's predictive capabilities and mitigate overfitting.
- Wrote a simple web interface using **Flask**, **HTML**, and **CSS** to visualize and interact with the system's predictions.

Clinic Management System | *Java, JUnit, Git* — June 2022 - August 2022

- Created a Clinic Management System adhering to **Clean Architecture** and **SOLID principles**, while incorporating **Design Patterns** for optimal code structure.
- Collaborated with a **team of 8**, leveraging **Git & Github** for efficient version control, pull requests, issue tracking, and code reviews.
- Ensured code reliability through comprehensive test coverage by developing robust test suites using the JUnit framework.

Technical Skills

Programming Languages: Python, Java, C, R, SQL
Frameworks/Libraries: PyTorch, scikit-learn, NumPy, FastAPI, Flask
Tools/Platforms: Git, Docker, Jupyter, GitHub
Databases: PostgreSQL, psycopg2